Design and Implementation of Big Data Crawling and Visualization System Based on COVID-19 Data

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***Abstract*—Based on the Scrapy framework, the system crawls the open COVID-19 data with Python crawler technology, and then stores it in the MYSQL database after data cleaning. Django framework is used as the front-end interface and react environment is used as the front-end framework to build the front-end of epidemic data. The covid-19 web page adopts a combination of map and visual graphics. With the help of rich visualization graphics library and map elements in Ant Design technology, the epidemic data are displayed in a diversified way, and the visualization of the new crown pneumonia epidemic data is realized. It plays a vital role in improving the awareness of self-protection and preventing epidemic disease.**

***Keywords*—COVID-19, Python crawler, Scrapy framework, MYSQL, Django, React, Ant Design, data visualization**

I. Introduction

Since covid-19 occurred, it is very valuable to visually display the global epidemic pattern through the visualization of epidemic data [1]. This has played a positive role in the traceability, monitoring and deployment of epidemic prevention and control. At the same time, it allows people to better grasp the trend of the global epidemic and enhance people's attention to the epidemic. The system uses data crawler technology to complete the crawling of epidemic data and vaccination data, and forms a data set after data processing. Data visualization technology is used to visualize the data set and realize the real-time update of epidemic situation. Help the public fully understand the latest epidemic data and the development trend of the epidemic.

II. System Introduction

The execution layer adopts the crawler technology under the Scrapy framework, which is a crawler framework based on Python 3.7.9 environment. The crawler is responsible for sending requests, data analysis and data decoding. And enter the MySQL database to complete the data storage. The presentation layer uses HTML, CSS, react framework and ant design framework to realize the visual display of page layout and epidemic information. Using JavaScript language and Django framework as the real-time data interface, obtain the real-time data of pneumonia epidemic, realize the data structure, and render the page data. The information list of pneumonia epidemic situation is shown in Figure 1, and the map of pneumonia epidemic situation is shown in Figure 3.

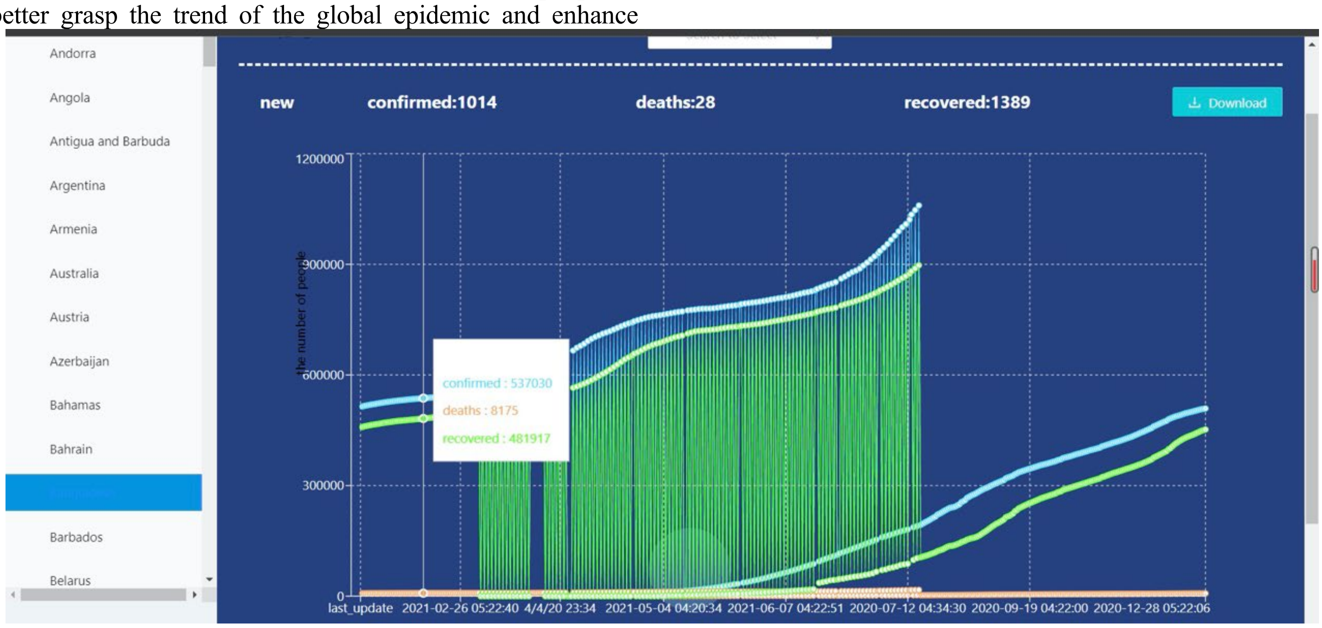

Fig.1 Cumulative epidemic trend chart

III. SYSTEM ARCHITECTURE AND IMPLEMENTATION PROCESS

A. System architecture

The system includes two parts: epidemic data crawling and processing, data visualization and analysis. Initiate the corresponding request according to the data requirements and receive the response from the server. The returned data is processed to form a data set [2]. Finally, the data set is visualized.

The system adopts four-tier architecture design, which is divided into resource layer, execution layer, function layer and presentation layer.

- The resource layer is dominated by relevant international health organizations and data centers, and climbs the main stations related to domestic mainstream epidemic situations at the same time.
- The execution layer adopts the crawler technology under the Scrapy framework, which is a crawler framework based on Python 3.7.9 environment. The crawler is responsible for sending requests, data analysis and data decoding.
- The function layer mainly involves two aspects: database and server. Here, MySQL 5.7 environment is used for data storage and data processing at the database level. Apache Web server is selected as the data processing mode based on memory.
- The presentation layer is mainly responsible for data visualization. This platform selects react based environment as the front-end building environment, and ant design as the data display UI framework to realize the visualization of epidemic data.

B. System implementation process

The system process is divided into the following steps:

1) Obtain the url and head information of the target web page;

2) send “request” to obtain response data;

3) Data storage and processing;

4) data visualization and data analysis.

The system operation flow chart is shown in Figure 2.

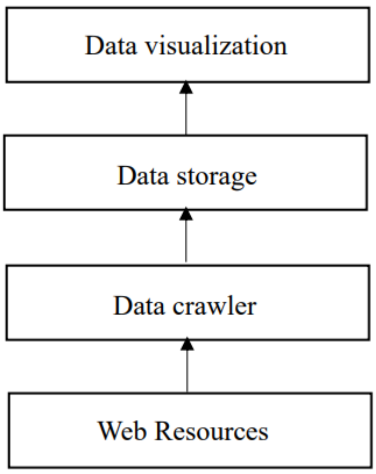

Fig.2 Architecture flow chart

IV. DATA CRAWLING AND PROCESSING

The system uses data crawler technology to obtain real-time epidemic data at home and abroad. The data comes from domestic public data resources.

Through the relevant URL data interface, the data required by all epidemic visualization systems are obtained and stored in the database.

The crawled data is cleaned, and then transformed into Jason's data dictionary. The description of the important key keys of the formed Jason data dictionary is shown in the Table I.

TABLE I. JASON DATA DICTIONARY

LastUpdateTime	Latest data update time
WorldTotal	Total world epidemic data
WorldAdd	New epidemic data compared with yesterday
AreaTree	National epidemic data
CurrentConfirmedCount	Current number of confirmed cases
ConfirmedCount	Cumulative number of confirmed cases
CuredCount	Number of cured and diagnosed
DeadCount	Number of confirmed deaths
NationDayList	National cumulative epidemic data in recent 60 days
NationAddList	National new epidemic data in recent 60 days

After data cleaning, conversion of Jason data dictionary, data extraction and other operations, the crawled data is divided into two data sets: global epidemic distribution data set and national epidemic trend data set.

Crawl the global epidemic cumulative data and global real-time epidemic data, and store them in global_ history_ In the URL, after the response data is obtained through the request and grouped, the global epidemic cumulative data and global real-time epidemic data can be obtained. After traversing and processing each group of data, a piece of data conforming to the database table structure can be found, and finally stored in the database.

V. DATA VISUALIZATION

A. Overview of visualization technology and platform

Visual analysis of data is a technology that comprehensively uses visual interface and analysis theory to help users interpret complex data. Visualization is an important way of data exploration.

In the context of big data, the mystery of massive data can be revealed only after it is reasonably interpreted and expressed. The form of visualization greatly improves the readability of data. People are no longer limited to observing and analyzing data information through relational data table, but also can see the data and its structural relationship in a more intuitive way. Generally speaking, the visualization of big data is complex and difficult to understand. It needs a series of processes such as extraction, cleaning, transformation and mining to display potential value information [3]. At present, big data visualization mainly includes text visualization, network (graph) visualization, spatio-temporal data visualization and multi-dimensional data visualization [4]. Rich and diverse visualization forms greatly facilitate people to obtain key information.

Nowadays, many countries and regions in the world have applied big data visualization technology to real life. For example, our covid-19 garden medical company has shown the massive new crown pneumonia epidemic data in form, and the netizens can get more comprehensive development of the epidemic situation, and change the situation of large amount of data and high redundancy, which provides convenience for users to view data.

However, covid-19 pneumonia charts on the website mostly contain only single elements, which can not directly reflect the difference between the outbreak trend and the data, and reduce the utilization of big data to some extent [5]. The system adopts the combination of map and statistical chart to reflect the change trend of national epidemic data through thermal map, two-dimensional histogram and dynamic broken line chart, and intuitively reflect the data differences under different indicators. At the same time, the draggable timeline will be used to dynamically visualize the epidemic data, vividly showing the content of the epidemic information big data and enriching the manifestations of the epidemic data.

Based on covid-19 pneumonia visualization platform, we need to select the appropriate off-line computing framework for big data. Apache Web server is selected here. Apache Web adopts the mode of data processing based on memory, which is a traditional web server. The service layer includes the web page framework and data services that the platform depends on. The Django framework is used to design the web service interaction scheme, and the back-end responds to the front-end request and transmits the data to the front-end in JSON format. The top layer is the application layer. Based on the classic modular front-end framework of react, the concise micro service development framework of react and the rich and diverse data visualization chart library and map element library in ant design UI, the global epidemic data visualization and epidemic data analysis visualization are finally realized.

B. Selection of visualization architecture technology

1) Echarts

Echarts is an open-source visualization library implemented by JavaScript, which can run smoothly on PC and mobile devices. It is compatible with most current browsers. The bottom layer relies on the vector graphics library zrender to provide intuitive, interactive and highly personalized data visualization charts, It has a wealth of visual chart types (including line chart, bar chart, map, and supports the mixing and matching between charts) [6]. It supports data sources in various formats such as two-dimensional table and key value, data block loading, asynchronous rendering, multi-dimensional data and rich visual coding methods.

2) Ant Design Architecture

Through ant design, the system reflects the changes of epidemic data in different time and space by means of broken line chart, histogram and world map, and based on the basic principles of comparative statistical chart method and zoning statistical chart method. In addition, the epidemic information of a specific country is displayed in the form of different symbol marks, two-dimensional coordinate maps of different heights and different colors. As shown in Figure 1, the broken line chart takes the time of epidemic development as the x-axis and the number of confirmed cases as the y-axis. A time axis that can be dragged is also attached below the X axis. Users can drag the time axis to obtain epidemic information in a specific time period.

The world epidemic map uses different colors to mark the cumulative situation of epidemic data in each region, from which we can intuitively see the differences affected by the epidemic in each region. The system adopts different visualization methods for different statistical data, which improves the intuitiveness, readability and spatial contrast of big data.

This platform selects ant design architecture for visual processing of data.

C. Visual content analysis

1) Visual content design

Covid-19 pneumonia covid-19, the new crown pneumonia epidemic data visualization platform was established on the basis of the national new crown pneumonia epidemic database. The original data comes from the existing data of relevant government departments and the World Health Organization. After data filtering, cleaning and integration, the detailed epidemic data of various countries and regions are obtained. The database includes global epidemic situation details database and user details database. The global epidemic situation details database includes the statistical epidemic situation data tables of various countries. Combined with the general requirements of database content and platform display, the system mainly visualizes the following two information objects.

(a) Epidemic distribution data of each province.

The main fields include country (region) name, country (region) geographical location, number of existing confirmed cases, etc.

(b) National epidemic data. The main fields include the number of confirmed cases, the cumulative number of deaths, and the cumulative number of cured cases.

2) Function design

(a) Global epidemic distribution map.

Covid-19 confirmed the map of the new crown pneumonia, as shown in Figure 3, with a grade of five, with a diagnosis of less than a few and a gradual increase in color. The figure shows the global epidemic distribution as of December 10, 2021. The rendering effect of the global epidemic distribution map is shown in the figure 3. The values of the current confirmed cases in the province are reflected by different colors, and the current confirmed cases in the country will be displayed when the mouse hovers. Users can intuitively see the global epidemic distribution. Global prevention and control efforts should be strengthened to prevent the secondary transmission of imported cases from abroad.

(b) National cumulative epidemic trend chart

The rendering effect of the national cumulative epidemic trend map is shown in the figure, with the date of nearly 10 months as the horizontal axis and the cumulative epidemic data as the vertical axis. Broken lines of different colors are used to distinguish the cumulative diagnosis, cumulative cure and cumulative death. When the mouse hovers, there will be the data display of the cumulative epidemic situation of the country on this day. As shown in the figure, the cumulative number of confirmed cases in Angola has increased significantly since March and by June 21, 2021. It can be seen that the epidemic situation in this country is growing rapidly at this stage, and there is great pressure on prevention and control.

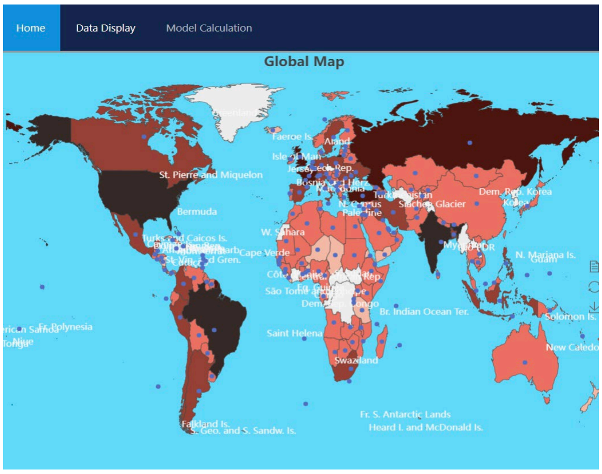

Fig.3 Global epidemic distribution map

VI. CONCLUISON

Big data visualization aims to enable users to have intuitive visual thinking on problems by integrating with maps, thermal maps, scatter charts and other chart forms, so that the public can "read and understand".

Based on the scrapy framework, the system crawls the open COVID-19 data with Python crawler technology, and then stores it in the MYSQL database after data cleaning. Django framework is used as the front-end interface and react environment is used as the front-end framework to build the front-end of epidemic data. The covid-19 web page adopts a combination of map and visual graphics. With the help of rich visualization graphics library and map elements in Ant Design technology, the epidemic data are displayed in a diversified way, and the visualization of the new crown pneumonia epidemic data is realized. It plays a vital role in improving the awareness of self-protection and preventing epidemic disease.

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